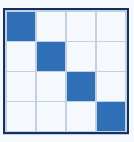
Eleven types and what each one forces

Each type is a condition on the entries, and the condition determines everything beside it. Grouped by what the condition constrains — the pattern of zeros, the transpose, or a power of the matrix.

1
GAPS

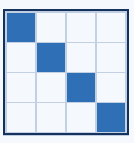
DEFINED BY A PATTERN OF ZEROS 5



Identity | $I_{ij} = 1$ if $i = j$, else 0

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
1	all 1	I	$AI = IA = A$

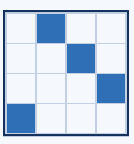
The multiplicative identity, and the one matrix that commutes with everything of its size. Every type below is a generalisation of it in some direction.



Diagonal | nonzero entries only on the main diagonal

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
$\prod d_i$	the d_i	$1/d_i$ entrywise	products and powers act entrywise

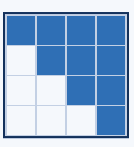
Everything is entrywise — which is the whole reason diagonalization is worth the effort of computing eigenvectors.



Permutation | exactly one 1 per row and column

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
± 1 — the sign of the permutation	roots of unity	P^{T}	reorders coordinates

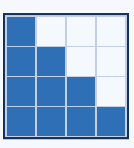
A zero pattern, not a power condition — you recognise one by looking at it. Orthogonal, so the inverse is the transpose, and it is the P of $PA = LU$, where it records the row swaps made during pivoting.



Upper triangular | all entries below the diagonal are 0

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
$\prod a_{ii}$	the a_{ii}	upper triangular	closed under products

The determinant and the spectrum are both read straight off the diagonal, which is what makes LU worth computing.

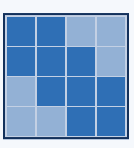


Lower triangular | all entries above the diagonal are 0

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
$\prod a_{ii}$	the a_{ii}	lower triangular	closed under products

The mirror image, and the L of an LU factorization. The transpose of one is the other.

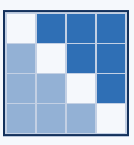
DEFINED BY A TRANSPOSE CONDITION 3



Symmetric | $A = A^{\mathsf{T}}$

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
not stated	all real	symmetric	$A = QDQ^{\mathsf{T}}$, Q orthogonal

The spectral theorem case: not merely diagonalizable but orthogonally so, which means P^{-1} is a transpose and nothing has to be inverted.



Skew-symmetric | $A^{\mathsf{T}} = -A$ — forces a zero diagonal

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
0 when n is odd	0 or $\pm bi$ in pairs	skew-symmetric	rank is always even

Setting $i = j$ in the condition gives $a_{ii} = -a_{ii}$, so the diagonal vanishes. Odd order then forces $\det = 0$, since the eigenvalues pair off and one is left over at zero.

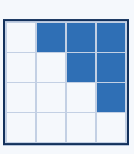


Orthogonal | $Q^{\mathsf{T}}Q = I$ — orthonormal columns

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
± 1	$\lvert\lambda\rvert = 1$	Q^{T}	preserves lengths and angles

The inverse is free — it is the transpose. This is what makes orthogonal factors numerically valuable: they cannot amplify error, since they change no lengths.

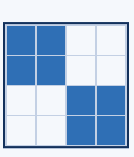
DEFINED BY A POWER CONDITION 3



Nilpotent | $A^k = O$ for some $k \geq 1$

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
0	all 0	none exists	$(I - A)^{-1} = I + A + \cdots + A^{k-1}$

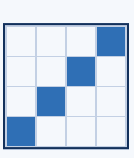
Never diagonalizable unless $A = O$: every eigenvalue is zero, so a diagonal form would be the zero matrix and hence so would A.



Idempotent | $A^2 = A$

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
0 or 1	0 or 1 only	only when $A = I$	$\operatorname{rank}(A) = \operatorname{tr}(A)$

A projection. Rank equals trace because the trace sums the eigenvalues and they are all zero or one — so it counts the ones.



Involutory | $A^2 = I$

DET	EIGENVALUES	INVERSE	KEY CONSEQUENCE
± 1	$+1$ or -1 only	itself	$A^{-1} = A$

Its own inverse. Reflections are the geometric case, and the eigenvalue -1 is the direction being reflected across.

Read down a column rather than across a row and the families appear. Every type defined by a power condition has eigenvalues confined to the roots of its own defining polynomial — $\{0\}$, $\{0, 1\}$, $\{\pm 1\}$ — because the condition is an equation the eigenvalues must satisfy. Every type in the zero-pattern group can be recognised by sight alone. The groupings are not filing — they predict the columns.